Ex.4 Time series

1. Find the partial ACF of the AR(2) process given by
   $X_t=(1/3)X_{t-1}+ (2/9)X_{t-2} + a_t$

2. Sketch the autocorrelation function of the process
   $y_t=-0.5y_{t-1}+a_t-0.8a_{t-1}$

3. The first eight values of the ACF and partial ACF of 60 observations on a quarterly economic index, and of the first differences, are shown below:

	Lag	1	2	3	4	5	6	7	8
X_t	r_k	0.95	0.91	0.87	0.82	0.79	0.74	0.7	0.67
	ϕ_{kk}	0.95	0.04	-0.05	0.07	0	0.07	-0.04	-0.02
∇X_t	r_k	0.02	0.08	0.12	0.05	-0.02	-0.05	-0.01	0.03
	ϕ_{kk}	0.02	0.08	0.06	0.03	-0.05	-0.06	-0.04	-0.02

Identify a model for the series. What can be done in order to make a better job of formulating a 'good' model?

4. The following tables presents the SAC, and SPAC of a data set of 150 observations. Tentatively identify a model describing this series.

Lag	1	2	3	4	5	6	7	8	9	10
SAC	-0.42	0.32	0.05	0.01	-0.11	0.11	-0.11	0.01	0.04	-0.04
SPAC	-0.42	0.18	0.29	0.07	-0.26	-0.07	0.04	0.03	0.04	-0.02

5. (a) Show that the following time series $\{X_t\}$ is stationary.

$$X_t = 0.7X_{t-1} - 0.1X_{t-2} + Z_t, \quad \{Z_t\} \sim NID(0, \sigma^2)$$

   (b) A stationary time series has a white noise representation, i.e.,

$$X_t = \sum_{j=0}^{\infty} \psi_j Z_{t-j}, \ \& \quad \sum_{j=0}^{\infty} |\psi_j| < \infty$$

   Using partial fraction or otherwise, find ψ_4 for the time series in (a).

   Hint: Use the backshift operator representation.

   (Ans: ψ_4=0.1031)

6. Consider the AR(2) process X_t satisfying

$$X_t - \phi X_{t-1} - \phi^2 X_{t-2} = a_t, a_t \sim NID(0, \sigma^2).$$

(i) For what values of ϕ is this a stationary process?

(ii) The following sample moments were observed:
$\hat{\gamma}(0) = 6.06, \hat{\rho}(1) = 0.687, \hat{\rho}(2) = 0.61$
Find estimates of ϕ & σ^2 by solving the Yule-Walker equations.

7. Derive the infinite moving average representation of the model

$$X_t = 0.6X_{t-1} + \varepsilon_t + 0.3\varepsilon_{t-1}, \varepsilon_t \sim WN(0, \sigma^2).$$

8. Suggest a plausible ARMA model for a series of 100 observations whose first ten sample autocorrelation coefficients are 0.61, 0.47, -0.05, 0.06, -0.21, 0.11,0.08, 0.05, 0.12, -0.01.